HINDE CLASSES

Std X Science — Model Paper Blueprint

Academic Year 2026–2027 • Based on Goa Board Grade 10 Assessment Scheme & Design of Question Paper

Chemistry	Physics	Biology

1. Weightage to Learning Outcomes

Learning Outcome	Marks	% of Marks
Remembering	21	30%
Understanding	21	30%
Application	17	24%
Analysis	4	6%
Evaluation	4	6%
Creativity	3	4%
Total	70	100%

2. Weightage to Content Units

Ch.	Unit	Tag	Marks
1	Chemical Reactions and Equations	Chemistry	4
2	Acids, Bases and Salts	Chemistry	6
3	Metals and Non-metals	Chemistry	4
4	Carbon and its Compounds	Chemistry	6
5	Life Processes	Biology	7
6	Control and Coordination	Biology	5
7	How do Organisms Reproduce?	Biology	6
8	Heredity	Biology	3
9	Light – Reflection and Refraction	Physics	6
10	Human Eye and Colourful World	Physics	6
11	Electricity	Physics	7
12	Magnetic Effect of Electric Current	Physics	6
13	Our Environment	Biology	4
	Total		70

3. Weightage to Form of Questions

Form of Question	Marks each	No. of Qs	Total Marks
Very Short Answer Type (VSA)	1 mk	18	18
Short Answer Type (SA-I)	2 mk	14	28
Short Answer Type (SA-II)	3 mk	4	12
Long Answer Type (LA)	4 mk	3	12
Total		39	70

4. Expected Time per Question Type

Form of Question	Time each	No. of Qs	Total Time
Very Short Answer Type (VSA)	3 min	18	54 min
Short Answer Type (SA-I)	5 min	14	70 min
Short Answer Type (SA-II)	7.5 min	4	30 min
Long Answer Type (LA)	12 min	3	36 min
Total		39	190 min

5. Difficulty Level

Easy	Average	Difficult
20%	60%	20%

Notes:
• 43 marks (Section A, 1 mark each × 18 + 2 Assertion-Reason) — 39 questions total, no overall choice, but internal choice (OR) is provided in 3 questions of 2 marks, 2 questions of 3 marks and 1 question of 4 marks.
• A separate 6 marks are reserved for diagram-based questions under the "Remembering" learning outcome — see the checklist on the next page.
• Final Board Exam = Theory 70 + Practicals 20 + Innovative/Assignment (internal) 10 = **100 marks**.
• **Chemistry 20 • Biology 25 • Physics 25** marks — no chapter deletions this year.

6. Question-wise Map (Model Question Paper, 2026–27)

Reconstructed from the Board's own model paper. Composite/OR questions are attributed to their primary chapter — use Section 2's table above as the authoritative chapter-wise mark total.

Q	Chapter	Type	Topic tested (short)
1	Chemical Reactions and Equations	VSA	Identify exothermic vs endothermic reaction
2	Electricity	VSA	Calculate resistance from V and I (Ohm's law)
3	Life Processes	VSA	Correct sequence of air passage during inhalation
4	Control and Coordination	VSA	Hormone preparing body for an emergency (adrenaline)
5	How do Organisms Reproduce?	VSA	Functions of labelled parts – female reproductive system diagram
6	Metals and Non-metals	VSA	Electrolytic refining of copper – identify correct statements
7	Electricity	VSA	Behaviour of current at the time of a short circuit
8	Acids, Bases and Salts	VSA	Product of calcium oxide reacting with water
9	Human Eye and Colourful World	VSA	Lens power to correct a given myopia defect
10	Acids, Bases and Salts	VSA	Relating pH value to H^{+} ion concentration
11	How do Organisms Reproduce?	VSA	Identify a bacterial sexually transmitted disease
12	Life Processes	VSA	Significance of transpiration in agriculture
13	Acids, Bases and Salts	VSA	Substance used to relieve indigestion (antacids)
14	Our Environment	VSA	Form of energy transferred between trophic levels
15	Metals and Non-metals	VSA	Protons and electrons in Mg^{2+}
16	Light – Reflection and Refraction	VSA	Sign of magnification → nature of image
17	Heredity	Assertion-Reason	Sex determination and the XX chromosome in women
18	Acids, Bases and Salts	Assertion-Reason	Sodium oxide as an amphoteric oxide
19	Chemical Reactions and Equations	SA-I	Balance the given equation; name the reaction type
20	Human Eye and Colourful World	SA-I	Coloured lights on towers; far point of a myopic pilot's eye
21	Light – Reflection and Refraction	SA-I	Ray diagram – concave mirror, object between F and C
22	Metals and Non-metals	SA-I	Why heating elements use alloys; ampere–coulomb relation
23	Electricity	SA-I	Advantages of AC over DC; frequency of AC supply in India
24	Acids, Bases and Salts	SA-I	How pH of the mouth influences tooth decay
25	Carbon and its Compounds	SA-I	Electron dot structure of methane; justifying alcohols
26	Chemical Reactions and Equations	SA-I	Compound used to disinfect swimming pools – name & formula
27	Life Processes	SA-I	Label parts of the digestive system diagram
28	Control and Coordination	SA-I	Hormonal response to sudden fear
29	Heredity	SA-I	Why bacterial division shows only minor differences; inherited traits
30	Control and Coordination	SA-I	Chemical coordination in plants / phototropism (or hormone functions)
31	Metals and Non-metals	SA-I	Reactivity series & electrolytic refining anode (or native/combined state metals)
32	Our Environment	SA-I	Construct a food chain; food chain vs food web (or trophic levels)
33	Life Processes	SA-II	Excretory system diagram; scientific reason for controlling urination

Q	Chapter	Type	Topic tested (short)
34	Human Eye and Colourful World	SA-II	Identify and correct a refractive defect of vision
35	Light – Reflection and Refraction	SA-II	Characteristics & use of a convex lens (or concave lens)
36	Magnetic Effect of Electric Current	SA-II	Magnetic field line pattern & properties (or electromagnetic induction)
37	Electricity	LA	Electrical energy calculation; labelled parallel-circuit diagram (or series/parallel lamps)
38	Carbon and its Compounds	LA	Cleaning action of soap – micelle formation
39	How do Organisms Reproduce?	LA	Modes of reproduction – rose/bryophyllum, unisexual flowers, pollen tube, spores

7. Diagram Evaluation Checklist

Every diagram below is explicitly listed by the Board as eligible for evaluation (6 marks reserved under ‘Remembering’). Prioritise these first when building diagram-based worksheets.

Chapter	Diagram
Chemical Reactions and Equations	Electrolysis of water
Metals and Non-metals	Electrolytic refining of copper
Metals and Non-metals	Action of steam on a metal
Carbon and its Compounds	Molecule of hydrogen; single/double/triple bonds (H■, O■, N■)
Carbon and its Compounds	Electron dot structures – methane, ethane
Carbon and its Compounds	Structures of cyclohexane and benzene; micelle formation
Life Processes	Open and closed stomata
Life Processes	Nutrition in Amoeba
Life Processes	Human alimentary canal; excretory system
Life Processes	Human respiratory system; structure of the heart
Life Processes	Transport & exchange of O■/CO■ (schematic)
Control and Coordination	Structure of a neuron; human brain; reflex arc
Control and Coordination	Endocrine glands in humans
How do Organisms Reproduce?	Binary fission (Amoeba); budding (Hydra)
How do Organisms Reproduce?	Bryophyllum buds; spore formation (Rhizopus)
How do Organisms Reproduce?	L.S. of a flower; pollen germination on stigma; germination
How do Organisms Reproduce?	Human male & female reproductive systems
How do Organisms Reproduce?	Regeneration in Planaria
Light – Reflection and Refraction	Concave & convex mirrors; ray diagrams (concave mirror, all cases)
Light – Reflection and Refraction	Image formation by a convex mirror
Light – Reflection and Refraction	Converging/diverging action of convex & concave lens
Light – Reflection and Refraction	Image position/size/nature – convex & concave lens (all cases)
Light – Reflection and Refraction	Refraction through a rectangular glass slab
Human Eye and Colourful World	Human eye; myopic/hypermetropic eye & correction
Human Eye and Colourful World	Dispersion of white light by a glass prism
Electricity	Circuit diagram & component symbols; circuit for Ohm's law
Electricity	Resistors in series; resistors in parallel
Magnetic Effect of Electric Current	Field lines around a magnet / straight current-carrying wire
Magnetic Effect of Electric Current	Fleming's left-hand & right-hand rules
Magnetic Effect of Electric Current	Electromagnetic induction between two coils
Our Environment	Food chain; trophic levels; food web; energy flow in an ecosystem

8. Portion & Calendar (2026–27)

Month	Chapter	Marks
April	Chemical Reactions and Equations	4
June	Control and Coordination	5
June	Electricity	7
June	Life Processes	7
July	Acids, Bases and Salts	6
July	Light – Reflection and Refraction	6
August	How do Organisms Reproduce?	6
September	Heredity	3
September	Magnetic Effect of Electric Current	6
November	Carbon and its Compounds	6
November	Metals and Non-metals	4
December	Human Eye and Colourful World	6
December	Our Environment	4